smallcase: The Strategy vs User

Explaining the gap between strategy performance and user implementation *

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Smallcase reports the performance of a strategy, while an investor experiences something more specific: money invested sequentially, rebalances may be executed later or at different prices, quantities are constrained to whole shares, and fees, taxes and dividends affect the eventual outcome. This project should answer: *How would the smallcase basket perform on my money? How did I actually perform? Why are those numbers different?*

I built a working product that reconstructs the strategy on the user's own cash flows, reconstructs the user's implementation from broker data, and attributes the gap through deterministic calculations. It then lets the user drill from portfolio-level performance into rebalance events, constituents and trades, including realised and unrealised P&L. An LLM sits only as an explainability layer over these verified outputs. The objective is to have a clearer way to understand what actually drove an investor's experience.

1 Problem & User Evidence

The starting observation was that while smallcase shows what an investor invested and earned, it does not show how their outcome compares with what the model strategy would have delivered on the same cash flows, nor provide a consolidated view of realised and unrealised performance across current and past constituents.

Public discussions reflected similar questions around return differences, rebalance timing, costs and taxes.[3] This was directional evidence rather than formal user research; the MVP was developed and tested on a real portfolio using smallcase's documented methodology and independently verified outputs.

The product opportunity was therefore to answer three linked questions: How would the strategy have performed on my money? How did I actually perform? And how can one explain the gap?

2 Problem Framing

I reframed the problem into four questions that build on one another:

1. *How did the strategy perform?* The published strategy/index return.
2. *How would it have performed on my money?* The strategy applied to my actual investment timeline and amounts.
3. *How did I actually perform?* Breakdown and trade level detail of my realised and unrealised performance or across current and past constituents.
4. *What explains the gap?* Differences arising from rebalance timing, execution prices, quantities and costs.

The key product distinction is the second question. A hypothetical one-time-investment smallcase return and an investor's actual return reflect on staggered cash-flows are not directly comparable. Reconstructing the strategy on the investor's own cash flows creates a like-for-like baseline and only then can the implementation gap be meaningfully measured and explained.

3 Product Hypothesis & MVP Scope

Hypothesis. If investors can compare their actual performance with what the strategy would have delivered on the same cash flows, and then drill into the causes of any gap, portfolio reporting becomes more diagnostic than a headline return or P&L view alone.

For the MVP, I deliberately constrained the integration rather than the underlying analysis: Zerodha was used as the broker, data was provided through Excel/CSV uploads instead of account-level API access, and the LLM was kept as an explainability layer over deterministic outputs rather than an autonomous agent

* Live product: strat-vs-user.streamlit.app. Data, code and methodology: github.com/....

with access to the portfolio. The objective was to validate the reconstruction, attribution and reconciliation on real data before expanding across brokers and workflows.

The product needed to:

1. create a like-for-like model return using the user's actual investment history;
2. reconstruct actual performance and reconcile it with broker-reported numbers;
3. explain differences at portfolio, rebalance, constituent and trade level;
4. consolidate realised/unrealised P&L, costs, dividends and tax-relevant gains;
5. support detailed natural-language questions without delegating financial calculations to the LLM.

I excluded recommendations, forecasting and judgments about whether an investor's execution was "good". The product explains what happened and why; it does not convert an ex-post outcome into investment advice.

4 Solution & Product Approach

The experience is intentionally simple: the first page projects the key outcome: the performance gap. Each subsequent page explains that result in progressively greater detail.

Front and center, the product compares the strategy on the user's cash flows with the user's actual return and quantifies the gap. That gap is attributed to different drivers: including execution price, timing, quantity, dividents and whole-share rounding effects. It lays out the underlying contribution by constituent, rebalance or trade. It also brings together realised and unrealised P&L across current and past holdings, along with all costs and dividends.

The product consolidates performance information otherwise spread across strategy history, broker statements and market data. The Q&A layer provides a natural-language interface over these deterministic outputs, enabling granular questions. For example, which rebalance contributed most to the gap or how realised gains were distributed across past holdings, without manually reconciling multiple sources.

5 Methodology

The calculation layer is deterministic. The smallcase timeline provides index values, rebalance dates, constituents and weights; Zerodha files provide executed trades, holdings, reported P&L and dividends; subscription costs are manually entered; and market prices are fetched only for the dates required by the analysis.

The engine keeps four views books:

- **Published strategy:** the smallcase index return, independent of the investor's cash-flow history.
- **Strategy on user cash flows:** the same strategy applied to the investor's actual investment dates and amounts, creating the like-for-like benchmark.
- **Reconstructed model portfolio:** a whole-share portfolio representing how those cash flows would have been implemented under the model strategy.
- **Actual portfolio:** performance reconstructed from the investor's executed trades and holdings

The analysis can distinguish differences arising from whole-share constraints, execution prices and quantities, costs and dividends, while reconciling back to the investor's reported P&L. Tax treatment is limited to identifying realised gains by holding period and asset class; personal tax rates are not estimated.

The LLM sits above, rather than inside, this calculation layer. It receives the computed outputs and uses them to answer granular questions across portfolios, rebalances, constituents and trades. Returns and gap attribution remain deterministic and auditable; the LLM's role is interpretation and explanation.

6 Validation & Reconciliation

The engine was checked against both ends of the system. The reconstructed smallcase view reproduces key Smallcase portfolio values; FIFO accounting reproduces Zerodha realised P&L; the event-level attribution identities hold numerically; per-stock contributions sum back to the portfolio-level effect; and the final waterfall closes with almost no unexplained residual. The index methodology was also independently replicated across historical transitions rather than fitted only to the final result.

Edge cases discovered during the build changed the implementation. Examples include distinguishing personal trades from smallcase order clusters, handling live-price rounding around whole-share quantities, avoiding stale market files around holidays, and flagging ambiguous rebalance pairing instead of silently choosing an answer. These were not cosmetic fixes: several materially changed attribution numbers and reinforced the need to expose assumptions rather than hide them.

7 Product Experience

The interface is intentionally minimal. The key outcome, the gap, takes the center stage, while each subsequent page gives more depth to that result. Detailed attribution is disclosed through rebalance, constituent and trade-level views, alongside realised and unrealised performance across current and past holdings. This keeps the headline experience simple without removing analytical depth.

The Q&A layer avoids adding a separate dashboard view for every possible question. Users can instead explore the underlying analysis conversationally.

8 Case Study: What the Tool Revealed

On the portfolio used to build and test the product, the headline strategy index returned **+1.34%**. Once the strategy was applied to the user's actual contribution dates, the like-for-like model return was only **+0.73%**: contribution timing alone changed the appropriate benchmark before any execution differences were considered.

The user's actual pre-cost return was **+4.58%**. The bridge between the model-on-user-money and the actual book was **₹85,240, or +3.85 percentage points**. Most of that difference came from the prices at which rebalances were actually executed relative to the model reference, with smaller reconciliation effects from whole shares, closing-price conventions and dividends.

The largest positive event came from a rebalance implemented eight days after the model date during a rising market. The tool reports that as an observed timing outcome, not as evidence of superior execution skill: the same delay in a falling market could have hurt performance. This distinction is important because the product is meant to improve understanding, not reward hindsight.

At constituent level, the same portfolio-level bridge can be decomposed into individual names, including stocks held in the past, while the broker book separately shows realised and unrealised P&L. This turns a single return gap into an inspectable history of what actually happened.

9 Limitations

The MVP makes several assumptions that remain visible to the user. Rebalance quantities cannot yet be independently reconstructed for every event because complete intraday portfolio valuations are unavailable; small order modifications can sometimes be indistinguishable from differences in the live price used by smallcase; and very delayed rebalancing can make event pairing ambiguous. Price conventions can also differ between broker statements and exchange data.

Uploaded tradebooks and P&L files are sensitive. The app processes them at runtime rather than storing them in the repository, but any production version would need stronger authentication, storage controls.

10 Next Steps

The next iterations would prioritise validating the product across more real-world use cases before adding more analytics:

1. **Test across more smallcases:** validate the reconstruction and attribution logic on different portfolio structures, rebalance patterns to identify additional edge cases.

2. **Benchmark on the same cash flows:** compare the investor's portfolio with relevant market indices using the same investment timeline and amounts, creating a more meaningful benchmark comparison.
3. **Broader ingestion:** extend beyond the current Zerodha-based workflow to support additional brokers and data formats once the core logic is validated across more examples.

11 Appendix: Calculation Notes

The complete engine walkthrough is available in the linked repository `here`. Each transition is represented by named effects that sum back to the final difference. Detailed formulas, event-level outputs, data fields, tests and known edge cases are also documented.

References

[1] smallcase. *Return Calculation Methodology*. `smallcase.com/meta/return-calculation`.

[2] smallcase Help. *Tracking & Managing smallcases*. Example help documentation describing index values, investment P&L and individual stock details. `iiflresearch.smallcase.com/help`.

[3] Public discussion used as directional evidence only: *Has anyone tried investing in smallcases?*, Reddit, September 2024. `reddit.com/r/IndianStockMarket/...`.

[4] Project repository. *The Strategy vs User*. `github.com/dhruwanda/strat-vs-user`.